unifying the five theories that I've so far neglected to mention. Just as a question's degree of difficulty can be changed radically by the translation from one string formulation to another, so, too, can the description of the geometrical form of spacetime. Here's what I mean.

Because string theory requires more than the three space dimensions and one time dimension of common experience, we were motivated in Chapters 12 and 13 to take up the question of where the extra dimensions might be hiding. The answer we found is that they may be curled up into a size that, so far, has eluded detection because it's smaller than we are able to probe experimentally. We also found that physics in our familiar big dimensions is dependent on the precise size and shape of the extra dimensions because their geometrical properties affect the vibrational patterns strings can execute. Good. Now for the part I left out.

The dictionary that translates questions posed in one string theory into different questions posed in another string theory *also translates the geometry of the extra dimensions in the first theory into a different extra-dimensionalgeometry in the second theory.* If, for example, you are studying the physical implications of, say, the Type IIA string theory with extra dimensions curled up into a particular size and shape, then every conclusion you reach can, at least in principle, be deduced by considering appropriately translated questions in, say, the Type IIB string theory. But the dictionary for carrying out the translation *demands* that the extra dimensions in the Type IIB string theory be curled up into a precise geometrical form that depends on—*but generally differs from*—the form given by the Type IIA theory. In short, a given string theory with curled-up dimensions in one geometrical form is equivalent to—is a translation of— another string theory with curled-up dimensions in a *different* geometrical form.

And the differences in spacetime geometry need not be minor. For example, if one of the extra dimensions of, say, the Type IIA string theory should be curled up into a circle, as in Figure 12.7, the translation dictionary shows that this is absolutely equivalent to the Type IIB string theory with one of its extra dimensions also curled up into a circle, but one whose radius is *inversely* proportional to the original. If one circle is tiny, the other is big, and vice versa—and yet there is absolutely no way to distinguish between the two geometries. (Expressing lengths as